Robustness of Turkish Punctuation Restoration under Informal Writing Noise

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Abstract

Punctuation restoration models can depend on orthographic cues that disappear in informal writing. We study Turkish casing loss, Turkish character folding, and internal character transpositions using deterministic perturbations that preserve word positions and punctuation targets. A published punctuation BERT provides an external reference. The controlled comparison uses clean and noise-augmented BERTurk models with identical initialization, examples, data order, optimizer settings, and three-epoch training budgets. A third training arm removes case loss without redistributing its noise budget. After validation-based development, we froze a held-out protocol comprising 122 model–condition evaluations. On 14,600 eligible test rows, augmentation changes clean punctuation macro-F1 from 84.3437% to 84.4824% and improves combined-heavy macro-F1 from 76.7502% to 80.5289%, a paired gain of 3.7787 ± 0.2191 percentage points across three corruption seeds. Removing case loss reduces matched case-loss and combined-medium F1 by 0.9585 and 0.8799 points, respectively. An exact-input-overlap sensitivity analysis preserves the main pattern. A subsequent instructor-feedback extension freezes corpus-anchored elongation, internal omission and partial folding outside the explicit training recipe. Its 52 additional evaluations yield Augmented minus Clean gains of 0.2491, 0.2290 and 0.1882 points, respectively. These are controlled distortions of the already scored test texts, not independent natural-text evaluation. One training seed, transported corpus anchors and unresolved dataset provenance limit broader conclusions.

1 Introduction

Punctuation restoration assigns marks to text whose punctuation is absent. It is often studied in speech transcription, but written inputs can also contain inconsistent casing, missing Turkish letters, and typing errors. Such edits retain the intended word sequence while changing surface evidence available to a cased Transformer. Turkish is a useful setting because locale-sensitive `I/ı/İ/i` casing and explicit Turkish character substitutions require care beyond generic English lowercasing.

We ask three questions: (1) which defined perturbations and doses reduce punctuation F1 most; (2) whether training augmentation improves robustness while maintaining useful clean-text performance; and (3) how punctuation classes and the case-loss training component contribute to the observed differences. The core comparison holds training initialization and budget fixed, rather than attributing differences from an independently trained published model to augmentation. Every planned held-out condition is reported, including small negative ablation contrasts.

The contribution is a *controlled Turkish writing-noise study*: deterministic word-level perturbations with exact target preservation, matched clean/augmented fine-tuning, a case-loss removal arm, and a reproducible held-out analysis with an overlap sensitivity cohort. We do not claim a new architecture or the first combination of punctuation restoration and noise augmentation. The project repository

Preprint.

provides source, frozen recipes, aggregate results, and the report; raw data and weights are retained separately.

After completing that original study, instructor feedback motivated a separate question: does the fixed augmentation recipe transfer to mechanisms outside its explicit operators when doses and morphology are anchored in informal Turkish? We report this follow-up separately in Section 6. The original report edition and all original metrics are preserved; the new results do not replace them or change any model.

2 Related work and design choices

Kurt and Çayır [1] release Turkish punctuation data and Transformer checkpoints. We use their pinned CSV release and published BERT as a reference, while independently specifying our training and evaluation recipes. BERTurk's cased encoder [2] initializes all controlled arms. Nagy et al. [3] study BERT punctuation restoration and report punctuation class metrics. We adopt first-subword token classification and emphasize punctuation-only macro-F1 with explicit treatment of non-punctuation positions.

Noise-aware punctuation/truecasing training has prior work: Sunkara et al. [4] use ASR hypotheses. Our spelling transformations preserve every word slot, so their ASR alignment and deletion rules are not imported. Belinkov and Bisk [5] and Karpukhin et al. [6] motivate examining character noise and augmentation, but their translation findings do not establish expected Turkish punctuation gains. Náplava et al. [7] motivate separating orthographic aspects; unlike their natural-error modeling, our original 2/5/10% doses are synthetic stress levels, not calibrated population frequencies. The supplemental study uses the Turkish train split of MultiLexNorm [11] for conditional corpus anchors and anonymous error-shape profiles. This transports finite-corpus evidence to our test domain; it does not establish representative Turkish usage rates.

CheckList [8] motivates condition-specific behavioral analysis. We retain the project's perturbation pipeline rather than use its toolkit. Lee et al. [9] motivate documenting overlap; our exact-token-list sensitivity filter is narrower than document-level deduplication. Dodge et al. [10] motivate the explicit distinction between training-seed variation and the corruption-seed variation measured here.

3 Data, task, and controlled perturbations

3.1 Release, eligibility, and held-out cohorts

We use the original train/validation/test CSV partitions from the pinned release.[1] Each word is labeled by the punctuation *after* it: `O`, `PERIOD`, `COMMA`, or `QUESTION_MARK`. A CSV row can contain several sentences and is not assumed to be a document boundary. The raw files contain 651 empty word slots with punctuation labels. Rather than delete targets or merge marks, the fixed `clean-v1` policy quarantines each affected row in full.

Table 1: Fixed eligibility policy. Quarantined rows remain locally preserved; the original CSV files are unchanged.

Split	Released rows	Quarantined rows	Eligible rows / words
Train	117,431	462	116,969 / 1,463,897
Validation	14,514	61	14,453 / 179,164
Test (primary)	14,661	61	14,600 / 181,485

The primary test cohort retains all eligible released test rows, including within-test duplicates. A second, fixed sensitivity cohort excludes a test row if its *original entire token list* exactly matches any released train or validation input. Case-sensitive token-list hashes use no labels, predictions, normalization, or length cutoff. The union excludes 155 rows: 148 match train, 90 match validation, and 83 match both. Sensitivity therefore contains 14,445 rows and 181,116 words. It filters the same

[1] Source revision: `7ede7a4ba9e51dad085a7f0d9b6c0a9c32622397`.

saved predictions; it is not a second independent test set. Missing document/talk identifiers leave document-level independence and near-duplicate separation unresolved.

Dataset licensing and the exact derivation/split provenance of these CSVs have not been established. The model card's license does not establish a dataset license. We publish aggregate measurements and source hashes, not the corpus, raw examples, or trained weights. Our filtered scores are not a direct reproduction of upstream unfiltered benchmark scores.

3.2 Noise operators and exact budgets

Each single-family condition changes exactly $K = \lfloor Np/100 \rfloor$ eligible word positions, where N is the number of original words in the full primary split and $p \in \{2, 5, 10\}$ denotes mild, medium, or heavy. The denominator includes words ineligible for that family. Thus test budgets are 3,629, 9,074, and 18,148 changed words per family. Stable SHA-256 rankings use original IDs, a split key, and corruption seeds 586/587/588; labels and model outputs do not enter selection. Severity masks are nested. Insufficient eligibility stops preparation rather than lowering the dose. The 10% maximum was chosen from train-only feasibility statistics; case-loss eligibility was the narrowest pool, about 13.35% of training words.

Case loss lowercases selected words with Turkish-specific `I`→`ı` and `İ`→`i` handling. **Diacritic loss** applies the explicit case-preserving map `ç,ğ,ı,ö,ş,ü`→`c,g,i,o,s,u` and its uppercase counterpart. This is a Turkish character-folding operator, not arbitrary Unicode accent removal. **Interior swap** exchanges one adjacent unequal pair strictly inside an eligible alphabetic word of length at least four, preserving its ends. Short words, digits, apostrophes, and unsupported characters are ineligible. This operator does not model all typos.

Combined reuses the three single-family plans in the fixed order swap, case loss, then folding. Overlap is audited, and its union coverage exceeds each component's budget: for seed586, combined-medium changes 14.3329% of test words and combined-heavy changes 27.3169%. It is a co-occurrence stress, not an equal-budget interaction estimate. Two separate full-loss endpoints transform every eligible casing or folding word; their test coverages are 13.4490% and 41.7495%, respectively. These endpoints are not additional 10% conditions.

Every perturbation preserves row/word IDs, word order, and original punctuation targets. Changed text is retokenized. The first subword receives the word's label; subsequent subwords, special tokens, and padding receive `-100`. No missing word or input longer than 512 subwords is silently dropped or truncated. All 39 held-out input views passed the recorded audits; the maximum prepared input length was 172, with no missing words or unknown subwords. Fixed intended targets do not prove semantic invariance for every corrupted spelling; ambiguity is part of the synthetic task's limitation.

4 Matched training and evaluation

4.1 Models and augmentation

Published is the released Turkish punctuation BERT, evaluated as provided.[2] Its training history and prior exposure to this test release are not controlled. **Clean**, **Augmented**, and **Without case loss** use the same saved BERTurk encoder/fresh four-class-head initialization at training seed 586. Their training rows and per-epoch row permutations are identical; each row appears exactly once per epoch.

In Augmented, exactly $\lfloor 0.5 \times 116{,}969 \rfloor = 58{,}484$ row IDs per epoch use a combined-medium view, replacing those rows in place. Remaining rows use their clean view; no extra training examples are appended. Word plans are first generated over the entire clean training split, then the selected row views are mixed. Consequently, "50% noisy views" is neither 50% corrupted words nor a guarantee of an edit in every selected row. Selected rows with no effective edit remain selected. Plans refresh once per epoch with distinct train-specific seeds. Realized mixed-view coverage is recorded in the input audits.

Without case loss uses the same initialization, row views, shuffle, remaining word plans, optimizer, and update budget while removing the case-loss operator. *The omitted budget is not redistributed.*

[2] Checkpoint revision: `dd8b9481ed3b3c05110d7e041ecbeb6a1c23b8b0`.

This estimates removal under the fixed recipe; reduced total corruption prevents an equal-dose claim about intrinsic component importance.

4.2 Optimization and checkpoint selection

All three arms fine-tune all parameters for three epochs / 10,968 optimizer updates. AdamW uses learning rate 2×10^{-5}, weight decay 0.01 except bias/LayerNorm parameters, betas $(0.9, 0.999)$, epsilon 10^{-8}, and gradient clipping at 1.0. Per-device batch 8 with four accumulation steps gives nominal effective batch 32; partial accumulation is flushed at epoch boundaries. Unweighted cross-entropy includes O and is normalized by valid word targets across each accumulation window. Dropout is 0.1; the schedule has 10% linear warmup followed by linear decay. Training uses FP32/eager attention on a T4.

Checkpoint selection uses highest clean-validation punctuation macro-F1 at completed epoch ends, choosing the earlier epoch on an exact tie. All arms selected epoch 3, so selected and final checkpoints coincide. No noisy-validation or test score selected a checkpoint. The recipe was fixed after validation development; no hyperparameter search, training-seed search, or early stopping was performed. Three epochs are the matched compute budget, not evidence of convergence or an optimal epoch count.

4.3 Frozen matrix, numerical controls, and metrics

Validation comprises three main models $\times 39 = 117$ evaluations, plus five new ablation evaluations paired with 15 existing references. After these validation results were inspected, the held-out protocol, models, cohort masks, conditions, and analysis rules were frozen before project test scoring. It is *validation-informed*, not a blind preregistration before development.

The held-out matrix contains 39 conditions for each main model: one clean anchor, 4 families $\times 3$ doses $\times 3$ corruption seeds, and two full-loss endpoints. The ablation has clean plus the four medium/seed586 conditions, matching its frozen validation scope. Total inference is $3 \times 39 + 5 = 122$ model–condition runs; primary and sensitivity reporting yield 244 score sets from the same predictions. There is no extra inference for sensitivity.

Inference uses the same Tesla T4 numerical policy across models and conditions: FP32, SDPA with the math backend, TF32 disabled, deterministic algorithms, batch 16, and four CPU threads. A shared clean-length/row-ID ordering fixes batch membership. This inference policy differs from training's eager attention; models are compared under the same recorded inference policy. Actual tensors, runtime/packages, sources, inputs, and eight validation-only readiness probes were checked before test scoring. All 122 conditions completed, followed by independent local recomputation of all 244 cohort scores from saved predictions and original word targets. No training or inference was repeated for analysis.

For punctuation class c, $F_{1,c} = 2TP_c/(2TP_c + FP_c + FN_c)$, with zero for a zero denominator. The primary score is $(F_{1,\mathrm{PERIOD}} + F_{1,\mathrm{COMMA}} + F_{1,\mathrm{QUESTION}})/3$. O positions remain in false-positive/negative counts and are omitted only from the punctuation averages. Macro/micro P/R/F1, class supports, and four-class confusions are also retained. Values below are percentages; differences are percentage points (pp). Mean $\pm$ sample SD describes the three *corruption* seeds only. Paired gains are computed per seed before aggregation, so the gain SD is not the difference between two marginal SDs. No p-values, confidence intervals, equivalence claims, or training-seed stability claims are made.

5 Results and analysis

5.1 Main held-out comparison

Table 2 reports every main-model condition. Clean-text macro-F1 is 84.3437% for Clean and 84.4824% for Augmented, a small descriptive $+0.1387$ pp difference. This observation does not establish clean-performance equivalence across training runs. Augmentation has positive mean gains in all listed family/dose groups (Figure 1), with the largest at combined-heavy: $76.7502\% \rightarrow 80.5289\%$ ($+3.7787 \pm 0.2191$ pp). The clean-trained model loses 7.5935 pp from its own clean

Table 2: Complete primary held-out matrix, punctuation macro-F1 (%). Stochastic conditions are means ± sample SD over three corruption seeds. Clean and full endpoints are deterministic. Δ is the paired Augmented minus Clean gain in pp. Ablation values are reported separately in Table 4.

Condition	Published	Clean	Augmented	Δ (pp)
Clean	84.0071	84.3437	84.4824	0.1387
Case loss, mild	83.0810 ± 0.1646	83.2890 ± 0.0821	83.8620 ± 0.0832	0.5730 ± 0.0935
Case loss, medium	81.5050 ± 0.1071	81.6460 ± 0.1313	82.8992 ± 0.1601	1.2532 ± 0.0505
Case loss, heavy	79.2674 ± 0.2516	79.1340 ± 0.2251	81.3246 ± 0.1641	2.1906 ± 0.0618
Diacritic loss, mild	83.9539 ± 0.0294	84.2402 ± 0.0265	84.4018 ± 0.0362	0.1617 ± 0.0521
Diacritic loss, medium	83.8259 ± 0.0420	84.1244 ± 0.0272	84.3583 ± 0.0273	0.2339 ± 0.0120
Diacritic loss, heavy	83.6073 ± 0.0365	84.0130 ± 0.0574	84.3013 ± 0.0579	0.2883 ± 0.1080
Interior swap, mild	83.6547 ± 0.0511	83.9855 ± 0.0521	84.3291 ± 0.0368	0.3436 ± 0.0371
Interior swap, medium	83.1434 ± 0.0503	83.4238 ± 0.0302	84.0301 ± 0.0204	0.6063 ± 0.0351
Interior swap, heavy	82.2406 ± 0.0339	82.4574 ± 0.1407	83.5796 ± 0.1157	1.1222 ± 0.1699
Combined, mild	82.6204 ± 0.1460	82.8196 ± 0.0709	83.6413 ± 0.0973	0.8217 ± 0.1358
Combined, medium	80.4421 ± 0.0866	80.4467 ± 0.0665	82.4459 ± 0.0919	1.9992 ± 0.0339
Combined, heavy	77.2021 ± 0.1631	76.7502 ± 0.3021	80.5289 ± 0.0906	3.7787 ± 0.2191
Full case loss	77.5536	77.3370	80.2493	2.9123
Full diacritic loss	82.8649	83.5346	83.7425	0.2079

Table 3: Primary per-class F1 (%). Heavy values average three corruption seeds; clean controls are single deterministic scores.

Input	Class	Clean F1	Augmented F1
Clean	PERIOD	96.1451	96.1348
Clean	COMMA	63.9560	63.9301
Clean	QUESTION_MARK	92.9301	93.3824
Combined, heavy	PERIOD	91.5565 ± 0.0862	92.4131 ± 0.0553
Combined, heavy	COMMA	51.6579 ± 0.1776	58.7611 ± 0.1188
Combined, heavy	QUESTION_MARK	87.0362 ± 0.6847	90.4125 ± 0.2312

anchor there, whereas Augmented loses 3.9535 pp. Robustness remains imperfect despite the improvement.

At matched 10% component word doses, Clean F1 is 79.1340% under case loss, 84.0130% under folding, and 82.4574% under swaps. Case loss is the most damaging *defined single-family condition* here; different eligible word pools and character changes prevent a universal ranking of noise types. Gains increase with severity: case-loss gains rise from 0.5730 to 2.1906 pp and swap gains from 0.3436 to 1.1222 pp. The combined curve is not an interaction estimate because its changed-word union is larger.

Published achieves clean F1 of 84.0071% and combined-heavy F1 of 77.2021%. It slightly exceeds Clean in combined-heavy and some other noisy conditions, despite lower clean F1. Thus a clean-only ranking hides robustness differences. This external comparison supplies context; the controlled augmentation estimate comes from Augmented versus Clean.

5.2 Class-specific behavior and contextual errors

Commas are the lowest-F1 punctuation class in both clean and combined-heavy text (Table 3). At combined-heavy, their F1 increases from 51.6579% to 58.7611% with augmentation, a 7.1033 pp gain; period and question-mark gains are 0.8565 and 3.3763 pp. Equal class weights make these comma errors important to macro-F1 even though question marks have much smaller support. Clean-test support is 15,465 periods, 12,164 commas, and 1,228 question marks; O has 152,628 positions. Full class P/R/F1 and confusion matrices remain in the public aggregate condition file.

The prelisted diagnostics compare each model to its own clean predictions. At case-loss-heavy/seed586, Clean creates 2,622 new word-class errors, including 2,270 at *unchanged words in edited rows*; Augmented creates 1,633, including 1,464 in that contextual category. At combined-heavy, corresponding totals are 4,047 and 2,455 (Table 9). All entirely unedited rows retain their

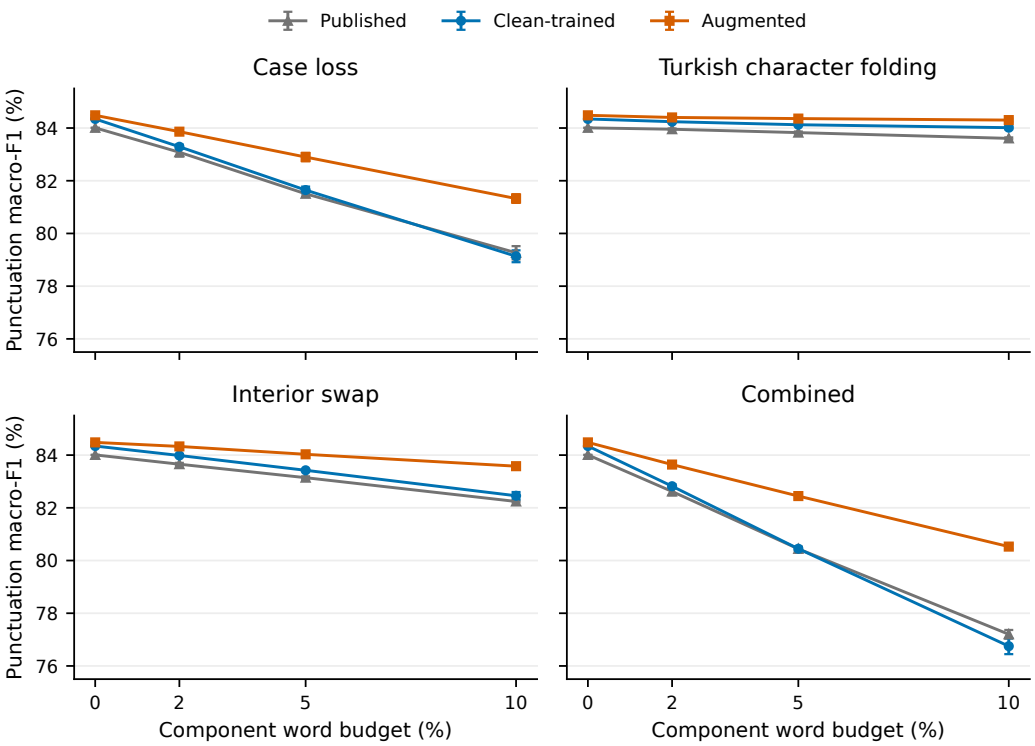


Figure 1: Primary held-out severity curves from verified aggregates. Error bars represent corruption-seed SD, not variation across training runs. Component doses in the combined panel have a larger union coverage.

Table 4: Matched primary ablation: corruption seed586 only. Δ is Augmented minus Without case loss, in pp. Both columns use the same seed; they must not be compared by subtracting a three-seed main-table mean.

Condition	Augmented	Without case	Δ (pp)
Clean	84.4824	84.3919	0.0905
Case loss, medium	83.0763	82.1178	0.9585
Diacritic loss, medium	84.3712	84.3917	-0.0205
Interior swap, medium	84.0256	84.1335	-0.1078
Combined, medium	82.4929	81.6129	0.8799

clean predictions in these checks. The counts show that edits can affect predictions beyond the edited word; they do not identify a causal mechanism inside attention or constitute punctuation-class FP/FN counts. A wrong punctuation substitution can contribute to more than one class metric.

5.3 Case-loss training ablation

Retaining case loss improves case-loss-medium F1 by 0.9585 pp and combined-medium F1 by 0.8799 pp (Table 4). The clean-test difference is $+0.0905$ pp. Small differences favor the removal arm under folding (-0.0205 pp) and swaps (-0.1078 pp). We report these unfavorable contrasts alongside the positive ones. The ablation's clean-validation advantage ($+0.1430$ pp for removal) does not recur on clean test, illustrating why validation and held-out roles should remain separate.

The component is useful against the matching perturbation under this recipe. Because removal also lowers total training noise, the experiment cannot show that casing is intrinsically more useful than another component at equal dose. Only five ablation conditions were planned in the original held-out study; no ablation severity sweep or additional training-seed confirmation was performed. The later

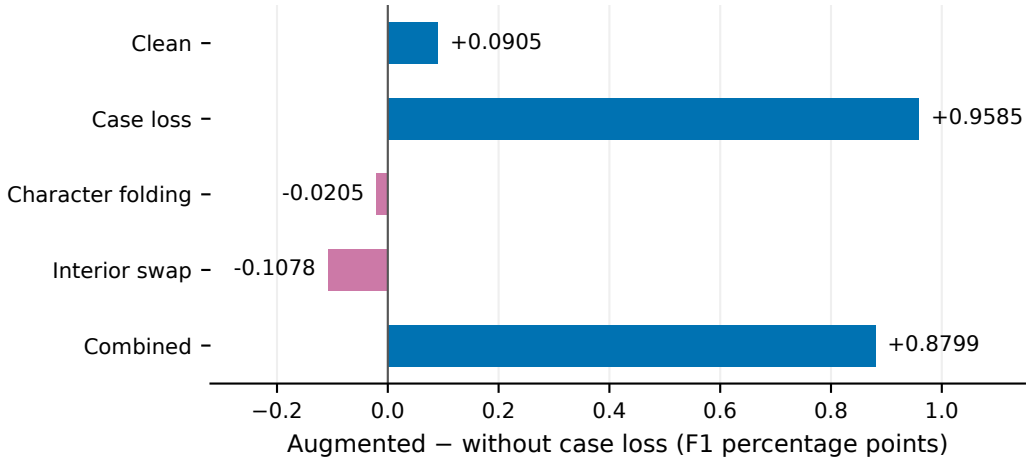


Figure 2: Fixed primary-cohort case-loss ablation contrasts. These are single-corruption-seed, single-training-seed descriptive differences.

transfer study evaluates the same removal checkpoint on its separately specified three-seed matrix, without extending the original noise grid.

5.4 Sensitivity and full-loss endpoints

Excluding the 155 exact train/validation input matches changes Clean/Augmented clean F1 to 84.2972%/84.4389%. Combined-heavy becomes 76.6662%/80.4613%, with a paired gain of 3.7951 ± 0.2178 pp. Case-loss and combined-medium ablation gains become 0.9637 and 0.8862 pp. The main descriptive pattern therefore survives this particular overlap filter (Appendix A); document-level independence remains unverified.

At full case loss, Clean/Augmented F1 is 77.3370%/80.2493% ($+2.9123$ pp). Full folding yields 83.5346%/83.7425% ($+0.2079$ pp). These endpoints differ substantially in coverage and information removed, and cannot be interpreted as a matched-dose family comparison. Even after augmentation, full casing loss still reduces performance relative to the clean anchor.

6 Corpus-anchored supplemental transfer

6.1 Train-only corpus evidence and fixed follow-up protocol

This extension follows instructor feedback and the original held-out analysis. Before new scores, we fixed the corpus profile, transformations, cohorts, models, contrasts and execution sources. It is a supplemental controlled follow-up on *already scored project test text*, not a new independent natural test or blind preregistration. No retraining, checkpoint selection or hyperparameter search uses these results. MultiLexNorm's test split remains unread; its lexical normalization targets are not treated as punctuation-restoration gold.

The official Turkish training file contains 570 tweet/sentence records and 6,443 aligned slots; 354 split/merge slots are analyzed separately. Restricting to one-to-one alphabetic Turkish/ASCII pairs permits target-preserving character operations. A broader short-word-inclusive analysis finds 89 elongations among 4,512 eligible opportunities, 50 one-character internal omissions among 4,174, and 43 partial folds among 621 words with multiple foldable occurrences. These denominators include unchanged eligible pairs and differ by mechanism. They must not be added or interpreted as percentages of all Turkish words. Minimum word lengths are two for elongation, three for omission and two for partial folding; partial folding additionally requires at least two foldable occurrences and the casing compatibility described below.

The original surface-preserving folding map has a Turkish `I/İ` case-order incompatibility at four corpus opportunities. Excluding those words and zero affected hits gives a separate *operational* partial

Table 5: Complete supplemental primary matrix, macro-F1 (%). Noisy cells are three-corruption-seed means ± sample SD. Without case uses the same selected removal checkpoint; no model is retrained.

Input	Published	Clean	Augmented	Without case
Clean anchor	84.0071	84.3437	84.4824	84.3919
Elongation	83.7501 ± 0.0200	84.1030 ± 0.0302	84.3520 ± 0.0756	84.2705 ± 0.0785
Internal omission	83.8678 ± 0.0219	84.1801 ± 0.0153	84.4092 ± 0.0681	84.2990 ± 0.0319
Partial folding	83.8992 ± 0.0290	84.2450 ± 0.0144	84.4332 ± 0.0559	84.3753 ± 0.0171
Matched full fold	83.9514 ± 0.0278	84.2854 ± 0.0467	84.4134 ± 0.0144	84.3815 ± 0.0303

rate of 43/617; the earlier case-neutral 43/621 result is retained unchanged. Per-tweet rate quartiles are all zero, so we use one empirical anchor per family rather than invent natural mild/medium/heavy levels. Uniform whole-split target-word sampling simplifies the corpus's observed tweet clustering.

For each family and seed, a new purpose-separated SHA-256 ranking selects exactly $\lfloor N_{\text{eligible}} h/o \rfloor$ clean word positions, where h/o is the relevant hit/opportunity ratio. On the primary test, elongation edits 3,355 of 170,103 eligible words, omission 1,826 of 152,483, and partial folding 1,803 of 25,881. The same selected words receive full folding as a seen-mechanism control. Sensitivity filters these primary realizations without reranking.

Anonymous *joint morphology templates*, rather than corpus vocabulary, determine edits. Elongation retains ordered relative run positions and added character counts: the 89 source events have median two extra characters, maximum 25, and 12 multi-run cases. Feasibility-conditioned templates map to distinct target character runs without clipping the tail. Omission deletes exactly one internal character, preserving both endpoints; 47 uniquely aligned events and three two-way ambiguous events contribute to a position distribution with uniform ambiguity marginalization. Partial folding preserves available count, changed count and relative sites jointly. Supported target counts use their conditional profile; unsupported counts map a donor fraction to a nonempty proper subset. This extrapolates 43/39/22 selected words at seeds586/587/588. Conditioning, mapping and extrapolation are declared approximations to the natural error process; exact rules and realized histograms are public.

Each of four fixed models receives a fresh clean anchor and four conditions at each of three corruption seeds: $4(1 + 4 \times 3) = 52$ forwards, or 104 cohort score sets. The original T4/FP32/SDPA-math/batch16 policy, clean-length batch membership, targets and first-subword alignment remain fixed. A separate runtime gate checks all four actual checkpoints and 24 validation-only technical probes before scoring. Saved-input checks and fresh tokenization retain every target, with maximum length 154, no truncation and no condition-specific row filtering. All saved predictions were independently rescored locally against original targets; all 104 score sets match. Each fresh clean anchor also exactly matches its original held-out counterpart in both cohorts.

6.2 Complete results and bounded transfer

Augmented exceeds Clean on average under all three new mechanisms (Table 5): $+0.2491 \pm 0.0843$ pp for elongation, $+0.2290 \pm 0.0635$ for omission and $+0.1882 \pm 0.0425$ for partial folding. The matched full-folding control has $+0.1279 \pm 0.0446$ pp. These low-dose differences are much smaller than the original combined-heavy gain; unlike that stress test, these rates condition on mechanism-specific eligible words. They are not a dose-matched severity comparison across studies or families. All paired values, sensitivity results and unfavorable class differences are retained (Table 6, Appendix E).

Absolute noisy performance includes the existing $+0.1387$ pp clean-anchor advantage. Own-anchor drops therefore provide a useful complementary view: Clean/Augmented lose 0.2408/0.1304 pp under elongation, 0.1636/0.0732 under omission and 0.0987/0.0492 under partial folding. This supports modest transfer under these specific edits, without a significance or training-stability claim. Retaining case loss gives small positive *absolute* noisy-score means over the removal arm (0.0816/0.1102/0.0579 pp). It does not uniformly reduce relative degradation: Without case loses only 0.1215 pp on elongation and 0.0166 on partial folding, less than Augmented. Thus this supplement does not establish that case-loss training specifically protects against all three mechanisms.

Table 6: Predefined supplemental paired differences (pp). P/S denote primary and sensitivity. Each difference precedes aggregation within seed.

Input	Aug−Clean (P)	Aug−Clean (S)	Aug−Without (P)
Clean anchor	0.1387	0.1417	0.0905
Elongation	0.2491 ± 0.0843	0.2521 ± 0.0845	0.0816 ± 0.0357
Internal omission	0.2290 ± 0.0635	0.2326 ± 0.0640	0.1102 ± 0.0665
Partial folding	0.1882 ± 0.0425	0.1913 ± 0.0427	0.0579 ± 0.0395
Matched full fold	0.1279 ± 0.0446	0.1305 ± 0.0448	0.0318 ± 0.0345

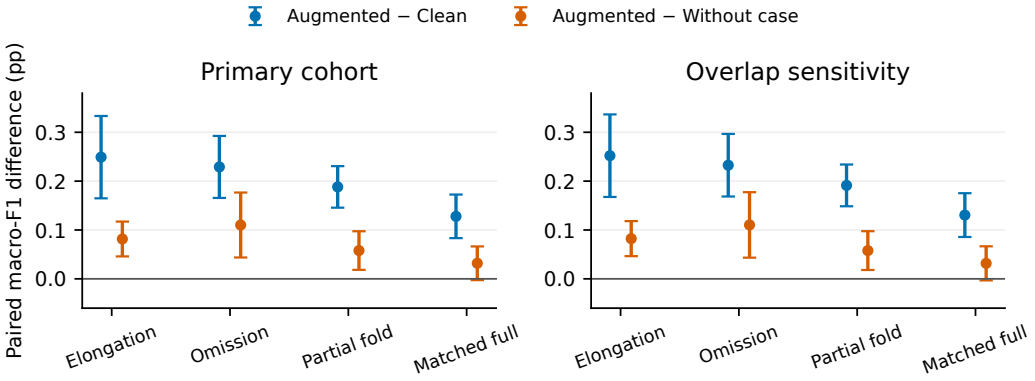


Figure 3: Supplemental paired contrasts. Error bars show corruption-seed SD, not confidence intervals or training uncertainty. Matched full is a control; character dose differs from partial folding.

Class behavior remains uneven: for example, Augmented minus Clean elongation F1 is negative for PERIOD (96.0856% versus 96.1093%) while positive for COMMA and QUESTION. QUESTION has only 1,228 primary targets, compared with 15,465 PERIOD and 12,164 COMMA. Appendix E reports all four models' class F1, supports, own-anchor drops and sensitivity matrix; full per-seed P/R/F1 and four-class confusions accompany the public results.

The partial-minus-matched-full control has mixed model signs: Published -0.0522 ± 0.0563 pp, Clean -0.0404 ± 0.0361, Augmented $+0.0198 \pm 0.0424$, and Without case -0.0062 ± 0.0368. Although both edit the same 1,803 words per seed, partial folding changes 2,171/2,188/2,140 characters and full folding 4,762/4,772/4,777. Consequently these contrasts cannot isolate whether a mechanism was explicitly seen during fine-tuning. No global pooled robustness score or post-score recipe refinement is introduced.

7 Limitations, responsible use, and conclusion

There is one training seed and one architecture. Three corruption seeds measure variation in perturbed inputs, not optimization uncertainty. The three-epoch budget does not establish convergence or the best achievable clean/robust trade-off. Word-preserving noise, including the new character insertions and deletions, still omits word-boundary changes, whole-word insertion/deletion, phonetic substitutions, broader keyboard errors, and code-switching. The original synthetic doses and transported finite-corpus anchors are not estimates of population Turkish usage. The corpus is small and selected; conditional morphology adaptation and uniform word sampling do not reproduce its full joint process. The same project test texts have been reused for the follow-up, so its controlled transfer results require unseen natural data for independent confirmation. The case-loss ablation changes both component presence and total noise exposure.

The released CSVs have unresolved licensing and exact split provenance. Exact input filtering does not remove all near duplicates or establish document separation; the published model's prior test exposure is unknown. Clean validation informed the recipe and checkpoint selection. The held-out protocol was frozen before our test scores, after development; future test-informed experiments

must be identified as exploratory and need new unseen data for strong independent confirmation. We make no state-of-the-art claim.

Restoration can change the interpretation of questions, boundaries, and pauses. These experiments do not validate high-stakes deployment or represent all Turkish speakers' writing. Human review remains appropriate when punctuation affects meaning. Public artifacts contain aggregate measurements and authored operator illustrations, while raw text, per-row examples/predictions, and weights are excluded.

Within these limits, matched noise augmentation improves Turkish punctuation robustness across the defined perturbation matrix with a small positive clean F1 difference. The strongest observed benefit is under combined-heavy noise; case-loss removal weakens matched casing/combined robustness. A separate corpus-anchored follow-up finds small positive mean Augmented–Clean differences for three mechanisms outside the explicit recipe, with uneven class effects and no uniform case-loss advantage in own-anchor degradation. Class and contextual diagnostics show where aggregate gains arise and where substantial errors remain. The result is a reproducible, bounded empirical study rather than a claim of universal robustness.

Acknowledgments and tool disclosure

The project uses the public data/model release by Kurt and Çayır and the BERTurk encoder. AI assistance (OpenAI Codex) supported implementation, debugging, literature synthesis, artifact verification, and report preparation. This disclosure does not assert an independently verified course AI-use policy. The NeurIPS 2026 preprint style is used for this course report; this document does not claim NeurIPS submission, acceptance, or peer review.

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A Complete overlap sensitivity matrix

The same primary predictions are filtered to the fixed sensitivity cohort: 14,445 rows / 181,116 words. No perturbation budget is resampled on this cohort. Values are punctuation macro-F1 (%); paired differences use the same corruption seed before mean/SD aggregation. Full per-class P/R/F1 is in the public CSV.

Table 7: All sensitivity conditions. Δ is Augmented minus Clean (pp).

Condition	Published	Clean	Augmented	Δ (pp)
Clean	83.9760	84.2972	84.4389	0.1417
Case loss, mild	83.0435 ± 0.1675	83.2359 ± 0.0830	83.8135 ± 0.0839	0.5776 ± 0.0941
Case loss, medium	81.4581 ± 0.1090	81.5810 ± 0.1341	82.8438 ± 0.1628	1.2628 ± 0.0513
Case loss, heavy	79.2066 ± 0.2552	79.0525 ± 0.2281	81.2574 ± 0.1674	2.2049 ± 0.0615
Diacritic loss, mild	83.9227 ± 0.0298	84.1933 ± 0.0266	84.3576 ± 0.0365	0.1643 ± 0.0526
Diacritic loss, medium	83.7959 ± 0.0422	84.0771 ± 0.0272	84.3137 ± 0.0276	0.2366 ± 0.0123
Diacritic loss, heavy	83.5764 ± 0.0372	83.9659 ± 0.0590	84.2565 ± 0.0586	0.2906 ± 0.1101
Interior swap, mild	83.6204 ± 0.0512	83.9366 ± 0.0536	84.2849 ± 0.0372	0.3483 ± 0.0387
Interior swap, medium	83.1064 ± 0.0494	83.3725 ± 0.0312	83.9843 ± 0.0204	0.6117 ± 0.0365
Interior swap, heavy	82.2034 ± 0.0340	82.4017 ± 0.1437	83.5321 ± 0.1179	1.1304 ± 0.1711
Combined, mild	82.5791 ± 0.1469	82.7653 ± 0.0711	83.5917 ± 0.0992	0.8264 ± 0.1351
Combined, medium	80.3899 ± 0.0892	80.3793 ± 0.0707	82.3892 ± 0.0932	2.0099 ± 0.0299
Combined, heavy	77.1358 ± 0.1730	76.6662 ± 0.3035	80.4613 ± 0.0919	3.7951 ± 0.2178
Full case loss	77.4783	77.2415	80.1728	2.9313
Full diacritic loss	82.8310	83.4868	83.6958	0.2090

B Realized coverage and diagnostic definitions

Table 8: Primary realized union/full-endpoint word coverage. Single-family 2/5/10% budgets use 3,629/9,074/18,148 words independently.

Input (seed586 where stochastic)	Changed words	Coverage
Combined, mild	10690	5.8903%
Combined, medium	26012	14.3329%
Combined, heavy	49576	27.3169%
Case loss full	24408	13.4490%
Diacritic loss full	75769	41.7495%

Table 9: New word-class errors relative to each model's own clean prediction, primary/seed586. Context means an unchanged word inside an edited row. All four reference classes, including O, enter these diagnostic counts.

Input	Model	Edited	Context	Unedited row
Case loss, heavy	Clean	352	2270	0
Case loss, heavy	Augmented	169	1464	0
Combined, heavy	Clean	1600	2447	0
Combined, heavy	Augmented	876	1579	0

For a given word, a new error means the clean prediction was correct and the noisy prediction is incorrect. Recovered errors are the reverse; changed errors are incorrect under both inputs with different predictions, and unchanged errors retain the same incorrect prediction. The fixed analysis also retains both-correct counts. Source-order example selection and diagnostic categories were set before test scoring; raw examples stay local and are not reproduced in this public report. The public diagnostic CSV contains every fixed model/condition/cohort/class partition, not only the new-error subset displayed here.

C Reproduction and evidence map

Read-only reproduction. The paper's tables can be rebuilt from public aggregate CSV/JSON files with `python scripts/build_report_tables.py`. This reads no dataset or model. `-check` verifies exact generated text and the source binding. Compile `report/main.tex` with the unchanged `neurips_2026.sty`; the report README documents the commands. Full experiments require the separately held inputs/models and their audited runtime, not merely this public repository.

Source-of-truth files. `docs/held-out-test-aggregates.csv` contains all aggregate metrics and paired contrasts; the condition-summary JSON contains supports and confusions. `docs/held-out-test-input-coverage.csv` records every input view's realized coverage; the diagnostic CSV records fixed outcome partitions. `configs/held-out-test.json` and `configs/held-out-execution.json` pin the protocol/model/runtime identities. Historical phase/status strings in frozen recipes are intentionally preserved; the current completed state is in `PROJECT_STATUS.md`.

Artifact identities. Full hexadecimal values distinguish the completed experiment from later report edits:

Run: `60e31867535de346accb64be45df04984434f6ff328b31c8819115cb689538a0`
Analysis: `e39f8803a6e6614363a6db97c1edce870757a2e909dec9dba5e93d0d9edc7f91`
Frozen protocol r2:
`ba6bf8decbd62a64f898a4eb47054776a97b3131fc8c6a20d8a50e22c6dc9653`
Local result ZIP:
`2463a23ed22dc9543341393bedfb902f693448330da860374c708748d4830925`

The verified local result archive contains 570 source members plus a manifest (16,011,054 bytes); training/model archives are separate. A later authorized Drive cleanup removed only backed copies after exact content/ID checks. Local results/models and retained selected Drive exports remain. Operational cleanup does not change any reported metric or experimental identity. Public Git contains aggregate reports, source, configuration, cleared notebooks, and this paper; data, weights, prediction text, run receipts, private course metadata, and build scratch files are excluded.

D Training history and compute context

Table 10: Original training-time clean-validation macro-F1 (%). Bold values are selected epoch exports, using the same prespecified F1 rule in each arm.

Epoch	Updates	Clean	Augmented	Without case
1	3656	83.2543	83.3313	83.1426
2	7312	83.4822	83.3883	83.5030
3	10968	**84.0365**	**83.9881**	**84.1310**

Every epoch evaluated the same 14,453 validation rows / 179,164 targets. Selection is by F1 rather than minimum validation cross-entropy: for example, Clean's epoch2 has lower validation loss than epoch3 but lower punctuation F1. The final-epoch selection does not show that additional epochs would not help.

Recorded training-loop times were 72.1855 minutes for Clean, 82.8746 for Augmented, and 94.75 for Without case loss. These include validation, checkpoint writes and model exports, excluding initial input/model checks. They are operational timings from separate sessions, not a controlled throughput benchmark. Held-out inference took approximately 77.8 minutes for all 122 conditions on the recorded T4 runtime; validation, preparation, calibration, analysis and archive work are separate. No energy estimate was measured. The report was assembled from existing evidence without new training or model inference.

E Complete supplemental evidence

The corpus-anchored extension preserves the original report edition and all original metrics. Values below are generated from the separate public transfer aggregates, not from rerunning a model. SD always refers to three corruption seeds and fixed trained models.

Table 11: Complete supplemental overlap-sensitivity matrix, macro-F1 (%). The same predictions are filtered to 14,445 rows /181,116 targets.

Input	Published	Clean	Augmented	Without case
Clean anchor	83.9760	84.2972	84.4389	84.3479
Elongation	83.7184 ± 0.0196	84.0554 ± 0.0306	84.3074 ± 0.0756	84.2252 ± 0.0782
Internal omission	83.8356 ± 0.0206	84.1325 ± 0.0138	84.3651 ± 0.0675	84.2547 ± 0.0320
Partial folding	83.8674 ± 0.0291	84.1981 ± 0.0144	84.3893 ± 0.0562	84.3315 ± 0.0173
Matched full fold	83.9200 ± 0.0282	84.2389 ± 0.0471	84.3694 ± 0.0143	84.3377 ± 0.0306

Table 12: All primary own-clean-anchor drops (pp). Positive means deterioration. Absolute noisy-score advantages and smaller own-anchor drops are distinct.

Input	Published	Clean	Augmented	Without case
Elongation	0.2570 ± 0.0200	0.2408 ± 0.0302	0.1304 ± 0.0756	0.1215 ± 0.0785
Internal omission	0.1393 ± 0.0219	0.1636 ± 0.0153	0.0732 ± 0.0681	0.0929 ± 0.0319
Partial folding	0.1079 ± 0.0290	0.0987 ± 0.0144	0.0492 ± 0.0559	0.0166 ± 0.0171
Matched full fold	0.0557 ± 0.0278	0.0583 ± 0.0467	0.0691 ± 0.0144	0.0104 ± 0.0303

Table 13: All models' partial-minus-word-mask-matched-full macro-F1 (pp). Word dose is matched; character dose is not.

Model	Primary (pp)	Sensitivity (pp)
Published	-0.0522 ± 0.0563	-0.0526 ± 0.0567
Clean	-0.0404 ± 0.0361	-0.0408 ± 0.0365
Augmented	0.0198 ± 0.0424	0.0200 ± 0.0428
Without case	-0.0062 ± 0.0368	-0.0062 ± 0.0371

Primary supports are invariant: O: 152,628; PERIOD: 15,465; COMMA: 12,164; QUESTION: 1,228. Sensitivity supports are O: 152,428; PERIOD: 15,325; COMMA: 12,152; QUESTION: 1,211. Table 14 reports class F1. All 104 sets of class precision/recall/F1/support and four-class confusion matrices, including O, are retained in `docs/transfer-evaluation-condition-summary.json`; the flat per-class CSV and aggregate CSV preserve every seed and specified contrast.

Table 14: Complete supplemental primary class F1 (%). Clean has one value; noisy conditions use mean ± sample SD. Rare QUESTION support is explicit above, and unfavorable PERIOD contrasts are retained.

Input	Model	PERIOD	COMMA	QUESTION
Clean anchor	Published	96.0549	63.2140	92.7524
Clean anchor	Clean	96.1451	63.9560	92.9301
Clean anchor	Augmented	96.1348	63.9301	93.3824
Clean anchor	Without case	96.1359	63.7663	93.2735
Elongation	Published	96.0136 ± 0.0071	62.6029 ± 0.1826	92.6338 ± 0.1331
Elongation	Clean	96.1093 ± 0.0099	63.3358 ± 0.1222	92.8638 ± 0.1172
Elongation	Augmented	96.0856 ± 0.0223	63.6261 ± 0.1286	93.3444 ± 0.2489
Elongation	Without case	96.1185 ± 0.0306	63.5136 ± 0.0717	93.1792 ± 0.1931
Internal omission	Published	96.0334 ± 0.0090	62.9216 ± 0.0445	92.6485 ± 0.0237
Internal omission	Clean	96.1164 ± 0.0077	63.6009 ± 0.0861	92.8231 ± 0.0442
Internal omission	Augmented	96.1266 ± 0.0023	63.7550 ± 0.1001	93.3460 ± 0.1112
Internal omission	Without case	96.1231 ± 0.0081	63.6072 ± 0.0574	93.1667 ± 0.0442
Partial folding	Published	96.0442 ± 0.0101	63.0241 ± 0.0738	92.6293 ± 0.0288
Partial folding	Clean	96.1257 ± 0.0032	63.7425 ± 0.0518	92.8669 ± 0.0171
Partial folding	Augmented	96.1174 ± 0.0035	63.8579 ± 0.1266	93.3242 ± 0.0869
Partial folding	Without case	96.1411 ± 0.0098	63.6678 ± 0.0230	93.3170 ± 0.0410
Matched full fold	Published	96.0401 ± 0.0097	63.0929 ± 0.0131	92.7213 ± 0.0617
Matched full fold	Clean	96.1309 ± 0.0111	63.7788 ± 0.0665	92.9466 ± 0.0643
Matched full fold	Augmented	96.1196 ± 0.0003	63.8326 ± 0.0593	93.2879 ± 0.0269
Matched full fold	Without case	96.1387 ± 0.0112	63.7034 ± 0.0194	93.3025 ± 0.0853

Reproduction. `scripts/analyze_transfer.py` validates frozen inputs, runtime/readiness/source bindings, all 52 condition commits and original targets, then recomputes both-cohort metrics and all paired contrasts from saved predictions on CPU. `scripts/report_transfer_results.py` publishes only aggregate evidence. The `-check` option of `scripts/build_transfer_report.py` verifies the separate paper tables and source binding without raw data or models; its `-figures` option redraws the supplemental plot. Original report-table inputs, tables, figures and run snapshots remain unchanged. No unit suite was run for this completion audit.

Supplement identities.

Protocol freeze: `b16cd070f37073b76503267668e27761922708eaa866ac3a80aab3fb790905d1`
Inputs: `235c0c01cf8b5a34b24e36f1bbdcfe8603c34e401171dd47ec9856c87a30b7a3`
Run: `8cf43b9d6a166baf005fb0320a52177f859c4b6215d06baaa0f93be57156d200`
Analysis: `7847a6b85dcf7d4f1296cff6495974bcbf4103c635b628fce2d8820a45ef06f6`

The verified recorded inference-loop sum is approximately 29.7 minutes; this is an operational timing including I/O, not a throughput benchmark. Preparation, readiness, downloading and analysis are separate. The results archive contains saved predictions/provenance and independent analysis; no weights are duplicated and no Drive cleanup is performed. The public original-study PDF is retained separately; private course copies and the exact prior source snapshot stay local.